NEXUS PEAR REV A

ESP32-S3 external antenna / remote AI / stereo I2S / documented SPI IPS

DO NOT ORDER: bench qualification and complete purchasing quote pending.

00_power



File: 00_power.kicad_sch

01_processor



File: 01_processor.kicad_sch

02_usb



File: 02_usb.kicad_sch

03_display



File: 03_display.kicad_sch

04_audio



File: 04_audio.kicad_sch

05_microphone



File: 05_microphone.kicad_sch

06_buttons



File: 06_buttons.kicad_sch

07_storage



File: 07_storage.kicad_sch

09_expansion



File: 09_expansion.kicad_sch

10_debug



File: 10_debug.kicad_sch

Read DESIGN_REVIEW_REVA.md before fabrication

Engineering prototype / DO NOT ORDER

Sheet: /

File: nexus-pear-reva.kicad_sch

Title: NEXUS PEAR REV A - System overview

Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 11/11

DO NOT ORDER — no smoke speedruns. Read the design review.

F1
3A / 24V PTC
VBUS
IN
OUT
+5V_FUSED

L1
3.9uH
SW_NODE
+3V3

Q1
2N7002 eFuse Inhibit
CC_HOT_3A
GND
EFUSE_EN

R40
332k
+5V
PGTH

C26
47n
SLEW
GND

C6
100n
+5V_FUSED
GND

C10
22u
+3V3
GND

D2
SMBJ10A VBUS TVS
+5V_FUSED
GND

R41
100k
PGTH
GND

U4
TPS25940LRVCR eFuse
GND
POWER_GOOD
PGTH
+5V
+5V
+5V
+5V
+5V_FUSED
+5V_FUSED
+5V_FUSED
DEVSLP
VGOOD
PGTH
OUT
OUT
OUT
OUT
OUT
IN
IN
IN
EN_UVLO
OVP
GND
ILIM
dVdT
IMON
FLT
EP
+5V_FUSED
+5V_FUSED
EFUSE_EN
OVP
GND
ILIM
SLEW
IMON
POWER_FAULT
GND

C11
22u
+3V3
GND

R35
34.8k
ILIM
GND

R42
10k
IMON
GND

#FLG1
PWR_FLAG
+5V_FUSED
SUPPLIED

R36
309k
+5V_FUSED
EFUSE_EN

R43
10k
+3V3
POWER_GOOD

C8
10u
+5V
GND

#FLG2
PWR_FLAG
GND
SUPPLIED

R37
100k
EFUSE_EN
GND

R44
10k
+3V3
POWER_FAULT

U5
AP63203WU-7 3V3 buck
+5V
+5V
+5V
FB
EN
VIN
GND
SW
BST
GND
SW_NODE
BST

#FLG3
PWR_FLAG
+3V3
SUPPLIED

R38
475k
+5V_FUSED
OVP

C20
100n
+5V_FUSED
GND

C9
100n
BST
SW_NODE

U8
TLV760330BZR always-on
+3V3_AON
+5V_FUSED
OUT
IN
GND
GND

R39
100k
+3V3
GND

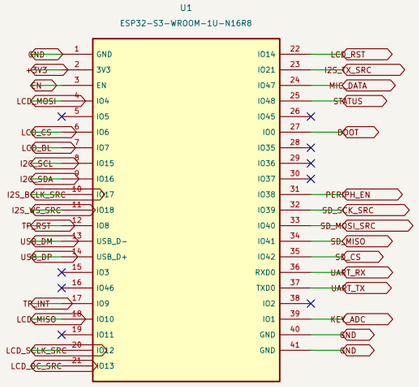
C21
1u
+3V3_AON
GND

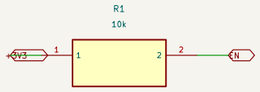
Read DESIGN_REVIEW_REVA.md before fabrication
Engineering prototype / DO NOT ORDER
Sheet: /00_power/
File: 00_power.kicad_sch

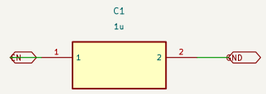
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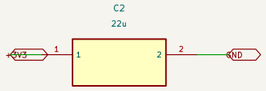
Size: A2	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 2/11

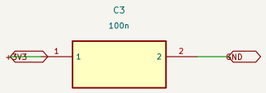
DO NOT ORDER — no smoke speedruns. Read the design review.

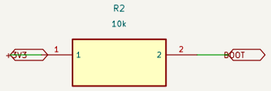












Read DESIGN_REVIEW_REVA.md before fabrication

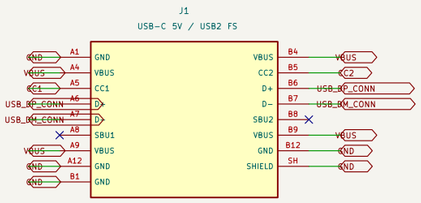
Engineering prototype / DO NOT ORDER

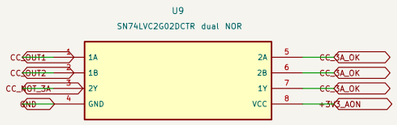
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File: 01_processor.kicad_sch

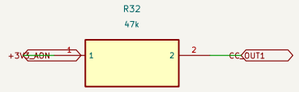
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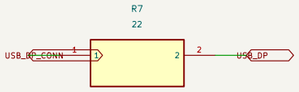
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 3/11

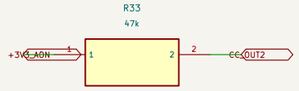
DO NOT ORDER — no smoke speedruns. Read the design review.

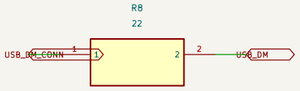


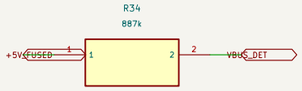


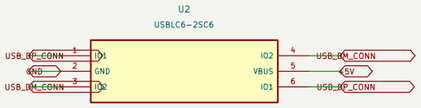


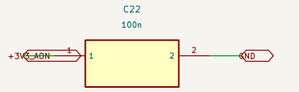


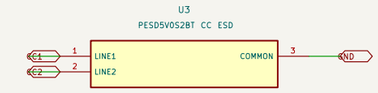


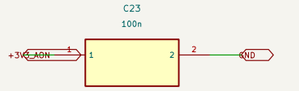


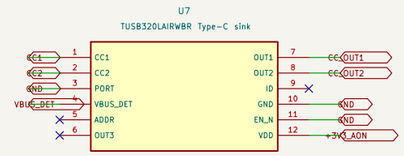












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Engineering prototype / DO NOT ORDER

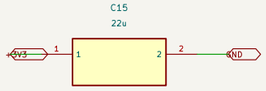
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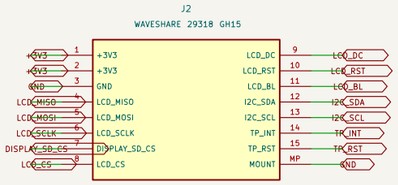
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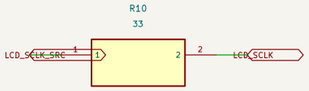
Title: NEXUS PEAR REV A — 02_usb

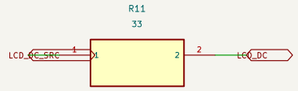
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 4/11

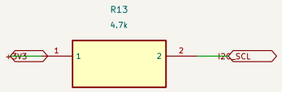
DO NOT ORDER — no smoke speedruns. Read the design review.

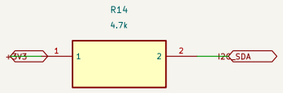


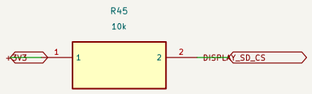












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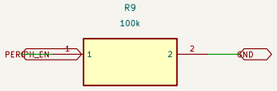
Engineering prototype / DO NOT ORDER

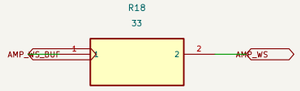
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File: 03_display.kicad_sch

Title: NEXUS PEAR REV A — 03_display

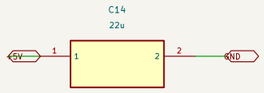
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 5/11

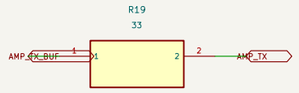
DO NOT ORDER — no smoke speedruns. Read the design review.

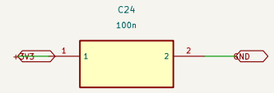


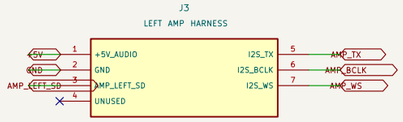


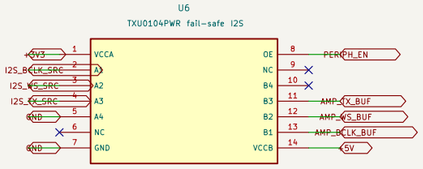


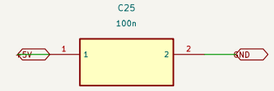


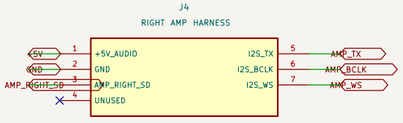


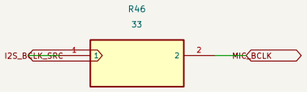


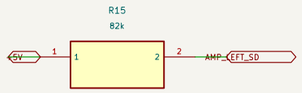


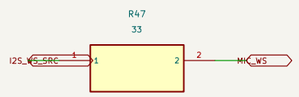


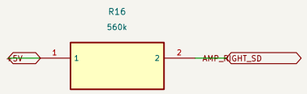




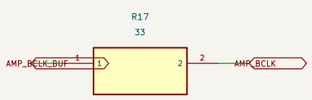


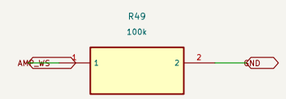












Read DESIGN_REVIEW_REVA.md before fabrication

Engineering prototype / DO NOT ORDER

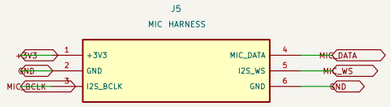
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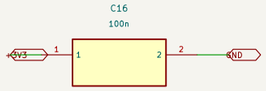
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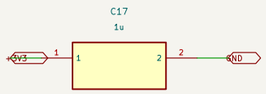
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Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
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DO NOT ORDER — no smoke speedruns. Read the design review.







Read DESIGN_REVIEW_REVA.md before fabrication

Engineering prototype / DO NOT ORDER

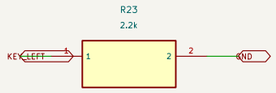
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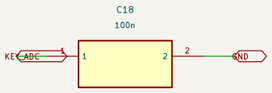
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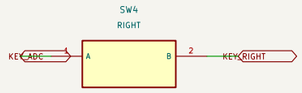
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 7/11

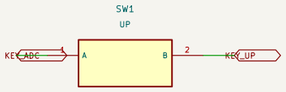
DO NOT ORDER — no smoke speedruns. Read the design review.

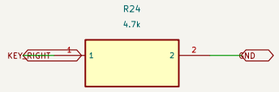


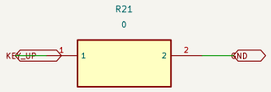


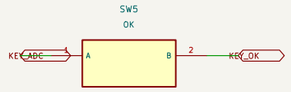


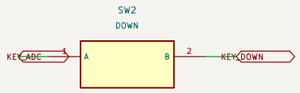


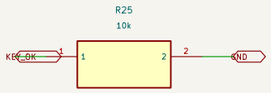




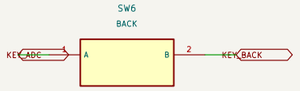


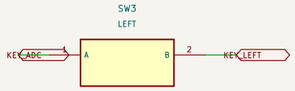


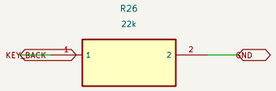












Read DESIGN_REVIEW_REVA.md before fabrication

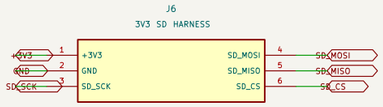
Engineering prototype / DO NOT ORDER

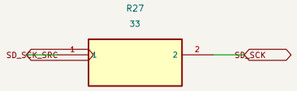
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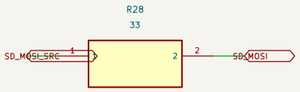
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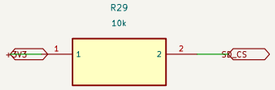
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 8/11

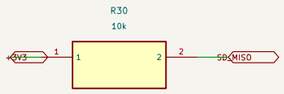
DO NOT ORDER — no smoke speedruns. Read the design review.

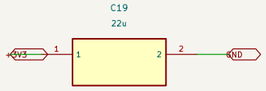












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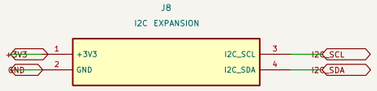
Engineering prototype / DO NOT ORDER

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Title: NEXUS PEAR REV A — 07_storage

Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 9/11

DO NOT ORDER — no smoke speedruns. Read the design review.



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Engineering prototype / DO NOT ORDER

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File: 09_expansion.kicad_sch

Title: NEXUS PEAR REV A — 09_expansion

Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 10/11

DO NOT ORDER — no smoke speedruns. Read the design review.

J7
UART_RECOVERY
GND
UART_TX
UART_RX
EN
BOOT
+3V3

SW7
BOOT

SW8
RESET

R31
1k

D1
STATUS_amber

TP1
VBUS

TP2
+5V

TP3
+3V3

TP4
GND

TP5
EN

TP6
BOOT

TP7
+3V3

TP8
+5V

TP9
I2C_SCL

TP10
I2C_SDA

TP11
MIC_BCLK

TP12
MIC_WS

TP13
MIC_DATA

TP14
SD_SCK

TP15
PERIPH_EN

TP16
CC_3A_OK

TP17
IMON

H1
M3_mounting

H2
M3_mounting

H3
M3_mounting

H4
M3_mounting

Read DESIGN_REVIEW_REVA.md before fabrication
Engineering prototype / DO NOT ORDER

Sheet: /10_debug/		
File: 10_debug.kicad_sch		
Title: NEXUS PEAR REV A — 10_debug		
Size: A3	Date: 2026-10-07	**Rev: A CANDIDATE**
KiCad E.D.A. 10.0.1		Id: 11/11